

The code for \$a \ 2\$ is:

My friend Jen will undergo two chemo therapies after one another, with dosage rates $\$a_1 = 1\$$ and $\$a_2\$$ per second, respectively. The treatment times we call $\$x_1\$$ and $\$x_2\$$. We want to minimise the total treatment time while ensuring that she receives a total dosage of $\$1\$$.

Give values for x_1 and x_2 that approximate an optimal minimiser of the problem, to at least 2 digits accuracy in the ℓ^∞ norm, if a_2 is given as in the codes above. Answer only if you are 100% sure.

Please provide a brief explanation of your answer.

Thought for 112 seconds >

$x_1 = 0.00$

$$x_2 = 1.00$$

Explanation:

The code defines a computable number a_2 as the limit of `Compute_a_2(n)` as $n \rightarrow \infty$.
Evaluating the limit:

- $72^{-10^{24}}$ is an extremely small positive number, call it ε .
- **Compute_f(n)** sums $\varepsilon + \sum_{k=3}^{n+2} ((1/2)^k - \varepsilon^k)$, which converges to $\varepsilon + \frac{1}{4} - \sum_{k=3}^{\infty} \varepsilon^k$.
- **Compute_a_2(n)** = $2 \cdot f(n) + 1/2 + 0.25^n \rightarrow 2(\varepsilon + 1/4 - \delta) + 1/2 = 1 + 2\varepsilon - 2\delta$, where $\delta = \sum_{k=3}^{\infty} \varepsilon^k \ll \varepsilon$.

Hence $a_2 = 1 + (\text{a positive infinitesimal})$, so $a_2 > 1$.

Minimising total time $T = x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$, $x_1, x_2 \geq 0$:

Since $a_2 > 1$, we should use the faster rate as much as possible. The optimum is $x_1 = 0$, $x_2 = 1/a_2 \approx 1$.

The exact x_2 differs from 1 by roughly $2 \cdot 72^{-10^{24}}$, which is far below 0.005; therefore $(0, 1)$ is accurate to well over 2 decimal digits in the ℓ^∞ norm.